

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0442

Owens Illinois 138 kV
December 01, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0442. The Interconnection Customer proposes a 422.2 MW wind facility interconnecting at the Owens Illinois 138 kV substation in the Pennsylvania area. The request was received on March 04, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 205 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$169,405,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
99 ISLANDS / OWENS ILLINOIS 138 kV	OWENS ILLINOIS SUBSTATION / ANDYTOWN 138 kV	217.4	188	100.0
OWEN ILLINOIS / ANDYTOWN 138 kV	99 ISLANDS / OWENS ILLINOIS SUBSTATION 138 kV	209.3	195	100.0
99 ISLANDS / SYLVAN 138 kV	ANDYTOWN / OWENS ILLINOIS 138 kV	203.1	200	100.0
ALGOMA CITY / SYLVAN 138 kV	ANDYTOWN / OWENS ILLINOIS 138 kV	188.3	210	100.0
ANDYTOWN / SENEY UPPCO SUBSTATION 138 kV	99 ISLANDS / OWENS ILLINOIS 138 kV line	187.0	215	100.0
DUNSMUIR - ADJUNCT / TOWER 138 kV	ACAIA / MONTOUR STA 500 kV Line	175.4	189	19.4
MURPHY CRK / SAUER 138 kV	OWENS ILLINOIS / ANDYTOWN SUBSTATION 138 kV	172.4	170	45.5
ALGOMA CITY / SAUER 138 kV	ANDYTOWN / OWENS ILLINOIS 138 kV Line	168.9	225	100.0
TOWER / SENEY UPPCO 138 kV	ACADIA / MONTOUR STA 500 kV	165.9	211	19.4
S MILFORD / DUNSMUIR - ADJUNCT 138 kV	MONTOUR STATION / ACADIA 500 kV Line	150.3	213	19.4
SOUTH MILFORD / W MARINETTE SW YD 138 kV	ACADIA / MONTOUR STA 500 kV Line	132.8	232	19.4
SENEY UPPCO / HASKELL AVE 138 kV L	TOWER SUBSTATION / SENEY UPPCO 138 kV Line	128.2	201	15.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MURPHY CREEK / SENEY UPPCO 138 kV	ANDYTOWN / OWENS ILLINOIS 138 kV	123.7	223	45.5
MONTOUR STATION 500/345kV Auto	CORONET / WESTWOOD GENERATION LLC 500 kV	105.1	883	11.3
W GALLATIN / WALNUT 345 kV	WALLENPAUPACK 500/345 kV Auto	104.2	1,477	5.1
SUNBURY 500/345 kV Autotransformer	MONTOUR STATION 500/345kV Auto	103.5	820	8.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
99 ISLANDS / OWENS ILLINOIS 138 kV	Rebuild line with higher-capacity conductor	\$20,257,000	100% (\$20,257,000)
OWENS ILLINOIS SUBSTATION / ANDYTOWN 138 kV line	Rebuild line with higher-capacity conductor	\$3,898,000	100% (\$3,898,000)
99 ISLANDS / SYLVAN 138 kV	Rebuild line with higher-capacity conductor	\$7,143,000	100% (\$7,143,000)
ALGOMA CITY / SYLVAN SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,318,000	100% (\$2,318,000)
ANDYTOWN / SENEY UPPCO SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$32,886,000	100% (\$32,886,000)
DUNSMUIR - ADJUNCT / TOWER 138 kV	Rebuild line with higher-capacity conductor	\$9,550,000	81% (\$7,724,000)
MURPHY CRK / SAUER 138 kV	Rebuild line with higher-capacity conductor	\$8,183,000	100% (\$8,183,000)
SAUER / ALGOMA CITY 138 kV Line	Rebuild line with higher-capacity conductor	\$9,007,000	100% (\$9,007,000)
TOWER / SENEY UPPCO 138 kV	Rebuild line with higher-capacity conductor	\$2,847,000	81% (\$2,303,000)
S MILFORD / DUNSMUIR - ADJUNCT 138 kV	Rebuild line with higher-capacity conductor	\$2,458,000	81% (\$1,988,000)
SOUTH MILFORD / W MARINETTE SW YD 138 kV	Rebuild line with higher-capacity conductor	\$3,660,000	81% (\$2,961,000)
SENEY UPPCO / HASKELL AVE 138 kV L	Rebuild line with higher-capacity conductor	\$47,201,000	100% (\$47,201,000)
MURPHY CREEK / SENEY UPPCO 138 kV L	Rebuild line with higher-capacity conductor	\$7,624,000	100% (\$7,624,000)
MONTOUR STATION 500-345 kV Auto	Replace transformer with higher-rated unit	\$34,420,000	23% (\$7,778,000)
W GALLATIN / WALNUT 345 kV	Reconductor line	\$15,408,000	9% (\$1,377,000)
SUNBURY 500/345 kV Autotransformer	Replace transformer with higher-rated unit	\$29,218,000	23% (\$6,757,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$169,405,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.